

Module 1 · Market Regime Identification

Market Data

$\tilde{r}_t, \sigma_t$



Regime Identification
(Gaussian HMM)



Bear

Bull

Sideways

→ r

▼ regime label r (per window)

Module 2 · Regime-Conditioned Diffusion Model

Real Windows

$\mathbf{x}_i$ (training set)



Learns $p(\mathbf{x} | r)$ — AdaLN Transformer

RevIN · DDPM · DDIM fast sampling



Candidate Synthetic Pool

Bear

Bull

Side.



Module 3 · Distribution-Aware Augmentation Allocation

sample w_r / N_{aug}

p_r^{tr}

Train dist.



p_r^{val}

Val dist.



$\delta_r = \max(p_r^{\text{val}} - p_r^{\text{tr}}, 0) \rightarrow w_r = \delta_r / \sum_{r'} \delta_{r'}$
allocate budget to underrepresented regimes



**Distribution-Aware
Synthetic Data**

Two-Stage Downstream Training

Real Data

$\{(\mathbf{x}_i, y_i)\}$



**Dist.-Aware
Synthetic**



R+S



Stage 1

Self-supervised Pretrain
(Real + Synthetic, no labels)



Stage 2

Supervised Fine-tune
(Real labels only)



**Return
Pred.**